

ENEL200-25S2 – Electrical and Computer Engineering Design

T4 Project Trailer Lights Tester

Group 09

Submission Date: 16th October 2025

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Executive Summary

This report presents the design and development of a trailer lights tester, capable of verifying the functionality of a trailer lights system by one person, without the need for a towing vehicle. New Zealand trailers are required by law to have operation rear lights. Traditional testing methods require both the connection of the trailer to another vehicle, along with two people to verify each light, which is inefficient for businesses managing multiple trailers. The proposed device is a portable, single person operable device that reduces labor requirements, saves time, and improves diagnostic accuracy.

The final design integrates electrical and software subsystems into compact durable housing, with an intuitive user interface. The design has four main subsystems: voltage regulation, load switching, fault detection and control logic. Power from the 18 V Makita battery is stepped down and regulated to 12 V using an LM25085 buck converter, paired with an external P-channel MOSFET. Switching regulator was chosen over linear due to its efficiency, reduced thermal loss and extended battery life. The converters passive network and component values were generated using TI Webench, and validated through PSpice simulation to ensure stable operation under a maximum 100W load.

Electronics are housed in a compact and portable 3D printed chassis, with durable 3mm walls. It uses M3 and M4 heat set inserts for secure mounting of PCB, lid, and Makita battery adapter, and uses cable glands for strain relieved cable entry. Flush mounted LCD display, momentary push button and a power switch on the front lid allows excellent user interface, without risk of damages from knocking the device.

The system is controlled using an Arduino nano microcontroller programmed in C as a state machine. It features single push button operation, LCD interfacing with the user, undervoltage battery protection, and automated fault finding algorithms. The software allows for easy future expansion to include other types of trailers and batteries.

The final design achieved all stakeholder and functional requirements, producing a safe, efficient and practical trailer lights tester. It provides significant time and labor savings for businesses managing trailer fleets, while being reliable and allowing repeatable testing.

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1. Introduction

In New Zealand it is a legal requirement to have working lights on back of a towed trailer if you intend to use it on the road [1]. Thus, it is essential that a trailer's lights are tested before use. This can be accomplished by connecting your vehicle's trailer plug to the trailer's socket and having one person operating the car's controls and another person standing behind the trailer observing the test. This technique is fine for a single trailer, but if you are operating a business with many trailers testing in this manner could become infeasible. It ties up two employees and requires you to move the vehicle each time you want to test a trailer. In addition, if a trailer's lights are functioning when you test them but fail once you connect a different vehicle, it is difficult to work out if the trailer is at fault or the vehicle. Having a handheld test unit which can be operated by a single person would save time and money for a business using multiple trailers and make it simple to differentiate between a fault on the trailer or on the vehicle. We anticipate that such a test unit could potentially see a wide range of use, as many companies hold large fleets of trailers. These could be equipment hire companies hiring trailers out to the public, hardware stores or service stations offering courtesy trailers, construction, landscaping, earthmoving and other trade companies which may have a large fleet of trailers, defense and emergency services and more. The primary stakeholder for this project is a Christchurch hardware store which employs one of the group members. They have a fleet of 10 courtesy trailers for customers to use and have requested a test unit to be developed. A wide scope was given to us in that the test unit only needs to illuminate the trailer's lights, be hand portable, and can be operated by a single person. This business has a range of power options available including a large selection of power tool batteries, 12-volt automotive batteries and chargers, as well as a large selection of primary cells. The trailers used by this business all accept a 7-pin trailer plug and feature a mixture of LED and incandescent bulbs.

2. Design Problem

This device should allow a single person to test the lights on a towed trailer which uses a 7-pin plug. The following image shows the pin-out of the plug (see Figure 1):

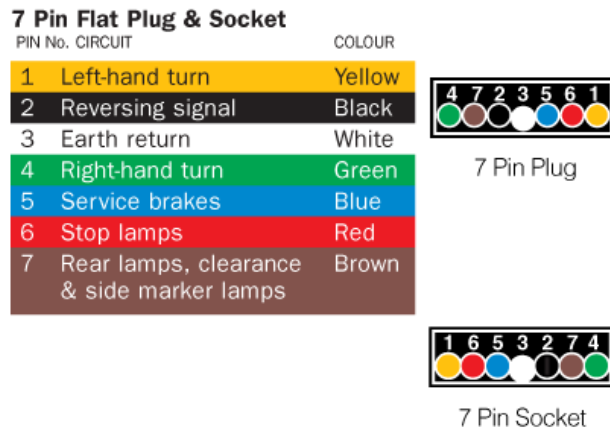


Figure 1: Pin-Out Table of Plug

The device should be hand portable, and capable of testing any type of 12-volt automotive trailer bulb including incandescent and LED bulbs. The device needs to test at least 10 trailers without requiring a recharge. To keep costs low the power source should be an 18-volt Makita power tool battery. This type of battery and its charging stations are widely available in the industry setting where this tester is to be employed. The battery should be simple to attach and remove from the tester. A summary of requirements and intended interfaces can be found in Table 1, and a block diagram of the system in Figure 2.

Table 1: Table of requirements & Stakeholders

Stakeholder	Requirement
Trailer light tester	R1 The trailer tester should test the lights on a 12-volt car trailer. R1.1 It must only require a single user to complete the test. R1.2 It must be hand portable. R1.3 It must connect to a 7-pin trailer plug.
	R2 The user must be able to visually verify if trailer lights are working. R2.1 The tester must illuminate each of the trailer's lights in sequence. R2.2 Trailer lights are lit long enough to quickly verify function. R2.3 The trailer lights must be lit brightly enough to see them in daylight.
	R3 The tester must indicate to the user if it has detected a fault in the trailer's lights.
	R4 The tester should be capable of testing 10 trailers without needing to recharge.
Hardware Store & Tester	R5 The tester must be powered by an 18-volt Makita power tool battery (type hardware store uses).

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	R5.1 The tester must indicate to the user if its battery requires recharging. R5.2 The tester should be inoperable if its battery requires recharging. R5.3 The battery should be simple to attach and remove.
	R6 Device components/electronics can be reliably accessed for repair R6.1 Device can be reliably taken apart without connections breaking/wearing out
	R7 Device is robust and will not easily break

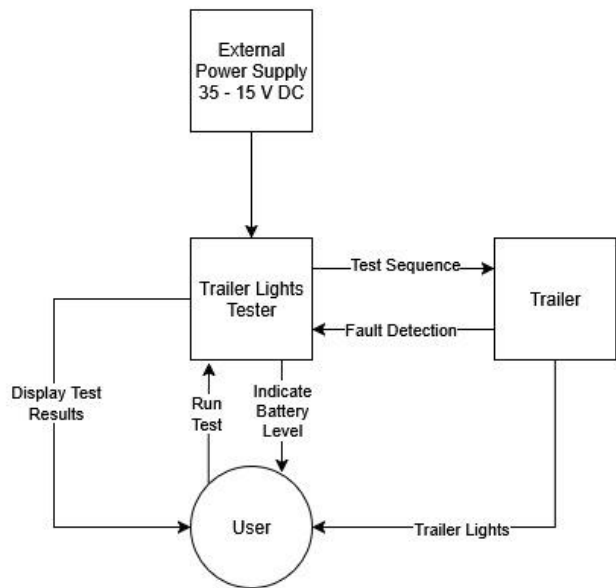


Figure 2: Trailer Lights Tester Block Diagram

3. Problem Definition

A standard seven pin connector is rated to provide up to 15 amps at 12 volts DC. Circuits for standard trailer lights like tail lights, blinkers, and brake lights have a capacity of about 10 amps. Thus our design must be able to provide at least 10 amps at 12 volts. As the power source is an 18 volt battery, a DC-DC converter or power supply may be required in the design. A typical incandescent automotive bulb found in a trailer will use 21 watts at 12 volts (1.75 amps), with LED lights using much less, on the order of 0.2 amps per bulb. Thus, a typical incandescent current draw for example, could be lighting both brake lights for a total of 3.5 amps.

Makita 18 volt batteries have a nominal voltage of 18 volts, a minimum operating voltage of 15 volts, and a maximum of 21 volts when fully charged. They come in capacities ranging from 1.6 Ah to 6 Ah.

Assuming that a 6 Ah battery is being used, a current draw of 10 amps per circuit, and an illuminated time of 1 second per circuit, ignoring losses and control electronics, we could test 360 trailers on a single charge. As shown above, many circuits will draw less than 10 amps and so we can consider this to be a lower bound on the number of trailers we could test.

4. Design Description

From Figure 3, the design consists of a buck converter, power sensing circuit, and 6 load switches that selectively power trailer light circuits, all controlled and monitored by an Arduino that also manages user interfaces.

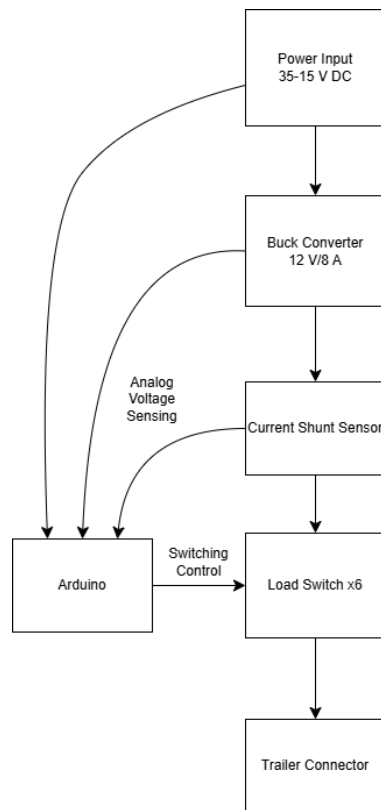


Figure 3: Block Diagram

4.1 Buck Converter

A buck converter (see Figure 4) is used to convert the wide input voltage range (15-35 V DC) down to a regulated 12V output, compatible with automotive systems. It was designed to supply a maximum of 8 A.

The design is based around the LM25085 buck controller IC. This drives a single P-FET high-side switch- in the production version a 3x3 DFN P-FET. A bank of ceramic capacitors (C9-C14) filters the supply voltage and supports a high transient current. The high-side switch feeds an LC filter formed by L1 and another bank of ceramic capacitors (C12, C13, C16), with additional bulk capacitance provided by an electrolytic capacitor (C7), which smooths the switching pulses into a near-steady 12V. Feedback from the 12V net is provided by a voltage divider formed by R23 and R24, switching is regulated to maintain the output voltage of this divider at the feedback regulation threshold of 1.25V.

A hard current limit is regulated by a 10 mOhm shunt resistor, R21. When the voltage across this shunt resistor exceeds a threshold set by R22, in this case 8 Amps, voltage is folded back to maintain 8 Amps.

D1 and D2 are Schottky diodes that form a path for freewheeling current to take when the P-FET is off, completing the buck circuit.

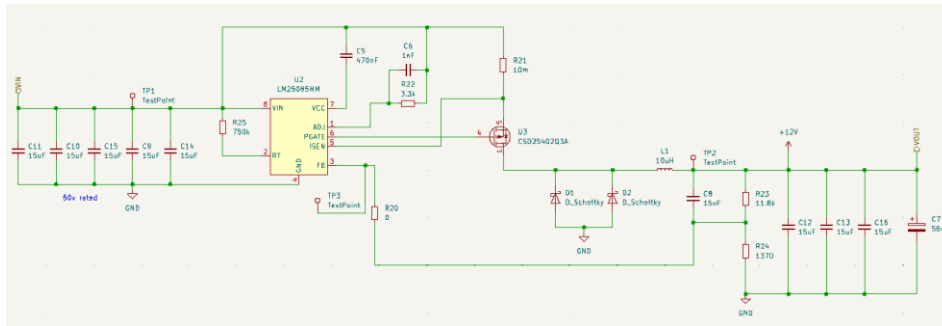


Figure 4: Buck Converter Schematic

4.2 Current Shunt Sensor

12V from the buck converter to the load switches is routed immediately through a current sense resistor (see Figure 5), R15. The voltage across this resistor is proportional to the current, according to Ohm's law. This differential voltage is connected to the inputs of a dedicated current sense amplifier, an INA181a1. This contains the entire trimmed and matched OpAmp feedback circuit, with a set gain of 20 for the a1 variant, so very little external circuitry is needed. With the 30 mOhm shunt fitted, the current analog output can range from 0 to 4.8 V, maximizing ADC range. An exception to the limited external circuitry is the differential low-pass filter formed by R13, R14, and C3. This ensures stable operation and output under noisy loads.

A pair of voltage dividers are placed at the input voltage and output voltage. The ratio of these was designed to maximize the ADC voltage input range of 0-5 Volts while still having enough headroom for safe operation. When voltage is measured from the input side, buck output voltage is measured, and buck output power can be calculated. Measuring voltage after the shunt is more indicative of the true power delivered to the load switches.

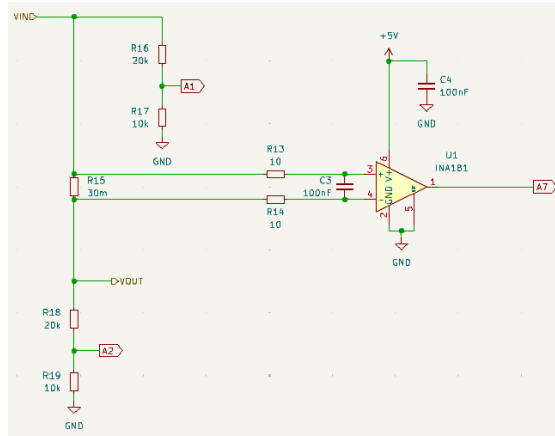


Figure 5: Current Shunt Sensor Schematic

4.3 Load Switches

The load switches are 6 identical units arranged in a bank (see Figure 6), one for each channel of a 7-pin trailer connector. Current is switched by a single TO-220 package P-FET per unit. In production these would be bolted to a single shared heatsink, like a copper busbar with large thermal inertia. The prototype used individual heatsinks. P-FET gate voltage is switched by an NPN BJT. When the base of this BJT is pulled low by the microcontroller, a pull-up resistor pulls the P-FET gate voltage to 12 V, giving it a VGS of effectively 0 and completely turning it off. When the NPN gate voltage is pulled to logic high, the NPN sinks this pullup current to ground, giving the P-FET an effective VGS of -12 and turning it completely on. This arrangement also has the benefit that logic high turns the P-FET on, and logic low turns it off, while usually P-FETs have inverted logic.

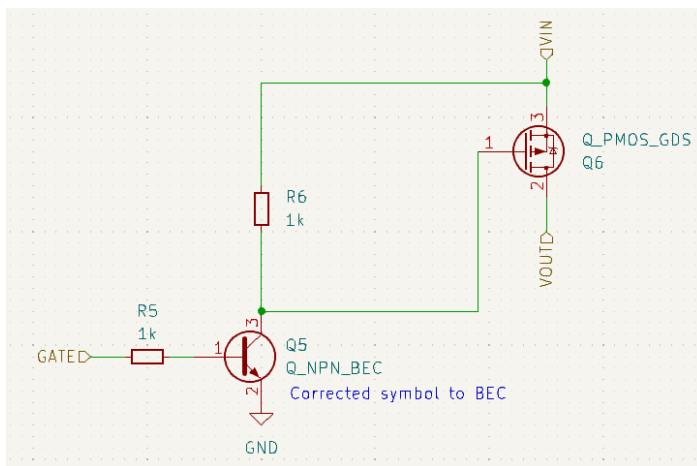


Figure 6: Load Switch Schematic

4.4 Software

The software running on the Arduino implements a state machine with states LOWBATT, IDLE, RUN, and RESULTS. Pressing the button on the device allows the user to move between states IDLE, RUN, and RESULTS. LOWBATT is entered if the battery voltage drops below 15v and locks the user out of leaving that state. Each of the 6 output pins is modeled using a struct which holds its pin number, pin name, fault state, and power level. The behavior of the software is shown in the following figure (see Figure 7), with the states highlighted in the colored boxes. Upon power up of the tester, the software first checks if the battery is over 15v. If so, it performs its initialization routine and moves to the IDLE state. If the battery voltage is less than 15v, it moves to the LOWBATT state. In IDLE, the device indicates to the user that it is ready to perform a test. From IDLE, pressing the button starts a 7 second delay after which it will move to the RUN state, lighting each trailer bulb in sequence for 1 second, with 0.5 seconds between each bulb. The 7 second delay is to allow time for the user to move to the back of the trailer to view the test. The bulb light timings are set to give the user enough time to see each light and be ready for the next light. A bulb is lit by setting its digital out pin to high. The results of the test are then displayed. If no faults are detected, pressing the button again will move the user back to the IDLE state. If faults are detected, the number of faults is displayed. Pressing the button will move the user to the RESULTS state, where each button press cycles the display through each fault. Holding the button down for 1.5 seconds counts as a button hold and moves the user back to IDLE. If during operation the battery voltage falls below 15v, the device will move to the LOWBATT state, no matter which state it is currently in. A visual overview of the software is provided in the following figure. The complete C code running on the device can be found in Appendix B.

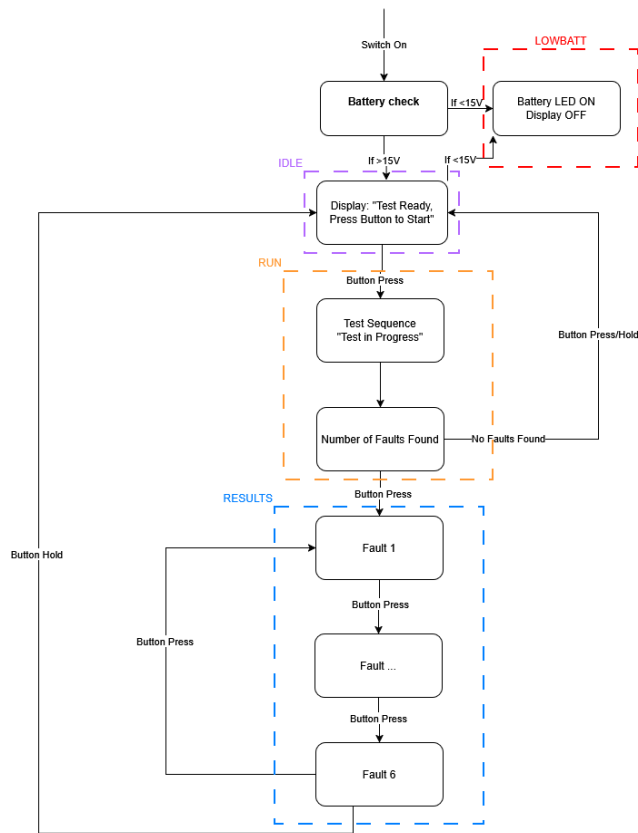


Figure 7: Software Block Diagram

5. Design Rationale

The overall design can be split largely into 4 smaller design problems; voltage regulation, load switching, fault detection, and control logic.

5.1 Voltage Regulation

The first design problem is voltage regulation. It is desirable for the design to be powered from Makita 18v batteries, which have a maximum voltage of 21v and a minimum of 15v. Incandescent bulbs might be able to withstand 21v for a few seconds, however LED “bulbs” are actually self-contained switching devices that would likely not tolerate voltages higher than the design 12v at all. Because of this, the input voltage range must be converted (and regulated to) 12v to ensure safe operation across all possible trailers. 2 main solutions were identified early in the design process; linear or switching regulation. Linear regulation was

immediately discarded; because this project is “safety critical” it is required that an actual human verifies that the trailer lights fully illuminate. Fully illuminating a circuit requires a maximum of 100 Watts- regulating that from 21 to 12 volts would exceed the power dissipation limits of nearly all linear power supplies, and the thermal dissipation of any reasonably sized tester device. Additionally, the specifications required that the tester would be able to test a reasonable number of trailers per battery charge; linear regulation would waste roughly a third to a half of total battery power, making it even more unsuitable.

A switching buck converter solves these issues, as they can regulate large output currents efficiently, allowing more trailers to be tested per battery charge and reducing the size of the thermal solution. In the worst case, up to 4 incandescent bulbs could be connected to a single circuit, dissipating a maximum of 100 watts when fully illuminated. This limits the choice of buck converter IC, because few controllers are rated for that power output, and practically no integrated converters are. Limiting the choice further was the short development time. If we chose a poorly documented or complex controller, we might not have had enough time to debug and develop the design in the required timeframe. These restrictions led us to select a buck controller with in-depth development documentation, the LM25085 controller. Again, as no integrated converter was suitable, this controller required an external p-channel MOSFET. MOSFET choice for the prototype was limited by the need to be hand-solderable. Out of through-hole packages, TO-220 are the most common, available from the store, and compatible with a wide range of heatsinks. Additionally, TO-220 P-FETs were available from the store with reasonably low RDS-on and gate capacitance, making their use a good choice for the buck converter. The “production” version was modified to use a 3x3mm DFN P-FET. SMD FETs generally have much lower gate charges than through-hole packages, and more modern FETs have vastly lower RDS(on) than those available from the store. These factors greatly reduce switching and resistive losses respectively, and in combination with the tight thermal coupling of SMD packages to copper pours this eliminates the requirement for an external heatsink.

Most of the component values of the buck converter were generated by Webench software and validated using Pspice, with a few key differences, such as the P-FET selection above. Aside from current sense resistors, catalogue through-hole resistors were deemed sufficient as most operated near DC. Webench suggested the use of 2x22uF input MLCCs and a single 47uF polymer output capacitor. Our design replaced these with 5x10uF input MLCCs and a combination of 3x10uF MLCCs and 56uF electrolytic capacitor at the output. This was necessary as no 22uF >=35v rated MLCCs were available, at the maximum input voltage 25v rated capacitors would derate significantly, potentially altering operation. Webench suggested the use of a 10-Amp rated TO-220 Schottky diode, this was replaced in the design with a single 5 Amp rated, store available diode. This was deemed sufficient as average diode current can be estimated by $I_D = I_{out}(1 - D)$, where Iout is at maximum 8 and duty cycle is at minimum 12/21, leading to an average diode current of ~3.4<5.

5.2 Load Switching

With a stable 12v supply, the next issue was to switch this 12v between circuits. By being able to select which circuit(s) are powered at a time, the instantaneous 12v power dissipation can be limited to the maximum per circuit power of 100w. Additionally, switching circuits on and off allows visual sequences to be played across the light circuits, making it easier for workers to identify faulty bulbs. Due to the amount of power being switched, the only solutions identified were MOSFETs or relays. Because of the >second timescale of switching, relays were viable however they were cost prohibitive and required additional circuitry to be controlled by an MCU and were thus discarded. Another option was N-channel MOSFETs, however due to the common ground shared between light circuits, low-side switching was impossible. This left P-channel

MOSFETs as the most viable solution, as they can switch current at the high-side. Fully turning off P-FETs requires VGS to be pulled close to 0, and fully turning on requires a VGS less than -10 for most FETs. Simply connecting the gate of the FETs to a GPIO pin would only allow the gate to be pulled to -7 VGS when high, and -12 when low, never fully turning off the FET. The solution for this is to pull up each gate to the source voltage and use a separate transistor to drain this current to ground. NPN transistors were chosen for this. NFETs are equally suitable, however NPN transistors were available for a lower BOM price. TO-220 package MOSFETs were chosen for their interchangeability and heatsink compatibility. The chosen IRF4905 MOSFETs of the “production” version have a safe operating area that encloses the maximum operating point of 12V 8A, and a very low RDS(on) of 20 mOhms. Although these have relatively high gate capacitance, due to the very long timescales of switching this doesn’t affect operation. The large reduction in resistive losses eliminates 6x heatsinks from the BOM cost compared to the prototype version.

5.3 Fault Finding

The next issue was fault finding. Incandescent bulbs have non-linear thermal resistance, simply measuring the resistance across each circuit with a low voltage would give an inconsistent result with the true operating resistance. Simultaneously, it would be impossible to determine LED bulb faults by measuring resistance as there is no guarantee that they would operate from a lower voltage or constant current source. This means that the only effective way to fault-find is to test the light circuits under normal operating conditions, which is from a 12V low impedance source. This limits the tests that could be run- an option that remained was measuring the flow of current from 12V through the circuit to ground. This metric can also be used to determine power, as finding the voltage across the light circuit is trivial using ADCs. Incandescent bulbs have nominal power of 21 W and LEDs are around 5 W. This large difference means any detected fault is more indicative than guaranteed. However, in the most common failure mode where bulbs fail open the dissipated power will be exactly 0, which is easily detectable. There were 2 main options to determine current: a shunt resistor and differential amplifier, or an integrated hall effect current sensor. Hall effect current sensors have lower series resistance however they are less accurate, and most importantly, are more expensive and not commonly available. Using a shunt resistor approach requires a shunt resistor with as high resistance as possible within power dissipation limits, to maximize the measurable signal. A large 2512 SMD package was chosen as this could dissipate 3W, which allowed an adequate shunt resistance of 30 mOhms at 8 Amps. Using a general purpose OpAmp would have been difficult as the common mode of the differential signal is huge at 12V; differentiating millivolts over/under this 12V common mode would require a feedback circuit with tightly binned resistors. The chosen solution to this was to use a commercial current sense amplifier, the INA181. This has laser-trimmed resistors in package, making it far easier to integrate.

5.4 Battery Protection

A 6A fuse is the first component from VIN. The Makita 18v battery is not designed to be discharged below 15v. Makita power tools and batteries have built in circuitry to prevent the battery from being damaged and will shut off the battery/tool around 15v. To mimic the protection found in the power tools, this tester uses a voltage divider (see Figure 8) to sense the battery voltage.

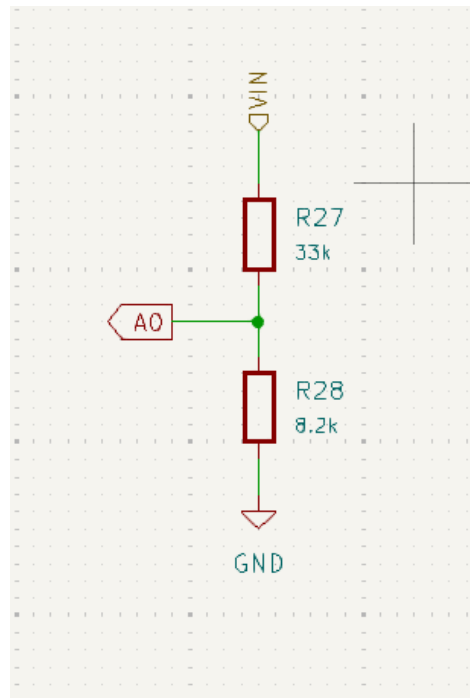


Figure 8: Voltage Divider Schematic

The divider ratio is designed so that a 25v input will be reduced down to 4.98v at the A0 input of the Arduino. This is to prevent damage to the arduino which can take a maximum of 5v on it's analog pins. While the maximum battery voltage expected is 21v, designing the input to be safe up to 25v was done to provide a margin of safety. According to Arduino documentation, for reliable readings, the output impedance of an analog signal source should be 10k Ohm or lower. The resistor values were chosen so that the output impedance of the divider is 6568 Ohms. In software the A0 pin ADC value is converted to a voltage. If the voltage falls below 15v, the software will move to the LOWBATT state as described previously.

5.6 Chassis

The chassis is composed of two main components: a bottom enclosure and a removable top lid (see Figure 9) that screw together using M3 screws, secured with heat set inserts in the bottom chassis, forming a compact cube-shaped housing. Both sections feature 5 mm filleted vertical corners to improve structural strength, reduce stress concentrations during 3D printing, and provide a clean external appearance that is comfortable to hold. Inside each corner, M3 heat-set inserts allow the lid to be securely fastened to the base, while additional raised cylindrical supports with M3 inserts enable the PCB shield to be mounted firmly within the enclosure. This design keeps the electronics stable and reduces vibration during handling or operation and allows adequate room for wiring and maintenance around the electronics.

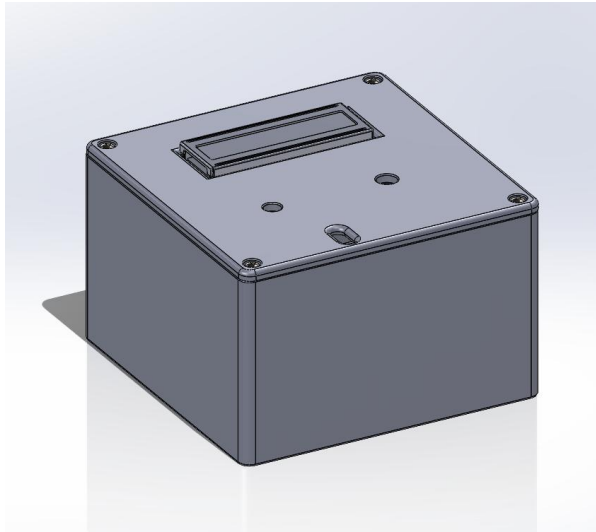


Figure 9: Assembly of full chassis & display

The underside of the chassis includes M4 heat set insert holes, designed to attach a Makita battery adapter, along with a hole in the bottom, to feed power wires from the Makita battery adapter. Major edges of the chassis are filleted to 2 mm, while smaller interior, along with screw holes and lid features have 0.5 mm fillets to enhance durability and 3D printability by avoiding sharp transitions that could lead to delamination or print defects.

The top lid complements the bottom enclosure and provides mounting points for the user interface components. It includes four corner screw holes that align precisely with the inserts in the chassis, ensuring a tight fit when assembled. The lid also features cutouts for a 16×2 I²C LCD display, including heat set inserts to hold it in place, a momentary push-button, and a power toggle switch. A small indicator hole allows the user to see the onboard LEDs located on the PCB shield during operation. Each screw recess is filleted by 0.5 mm, allowing screws to seat neatly and preventing cracking around the holes during repeated assembly.

The chassis was developed through an iterative process, balancing structural robustness with manufacturability and functionality. Early ideas explored rectangular and cylindrical chassis shapes, with a battery housing. These early designs were developed around the exported STEP file from the PCB shield design on Kicad (see Figure 10). After comparing alternatives, such as print practicality, component fit and assembly, a 130 mm x 130 mm cube was chosen, as it most efficiently fits the 100 mm x 100 mm PCB board with room to wire, and is most practical to print.

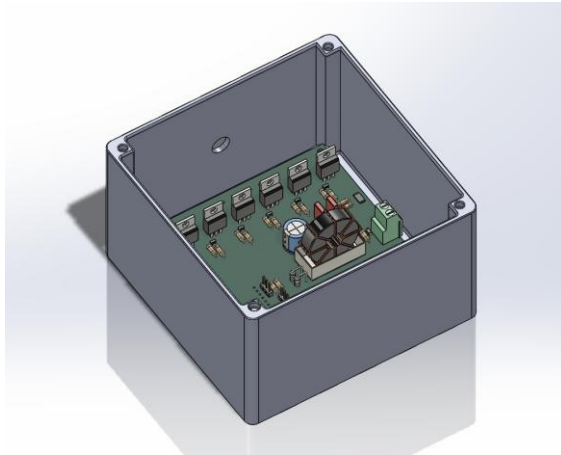


Figure 10: Bottom Chassis with PCB shield

6. Evaluation

6.1 Electronics

The efficacy of the buck converter and load switch solutions was tested through DC load testing, with values between 1 and 60 Watts. At loads of 25 Watts or under the load switches and buck converter worked as expected, even under long term (~5 minute) continuous tests which are beyond the design requirements. However, as delivered power increased the converter switch, and especially the load switch P-FETs, began to be thermally constrained. While the heatsinks were noticeably warm at 25 W, they heated exponentially with increasing power, and at 50 W the load switch P-FET failed. From the 25 W test parameters (Figure 11), it is apparent that the power path drops a considerable amount of voltage, around 0.4 V, even under a small current of 2.2 A. This power is mostly dissipated in the load switches, which in the prototype used P-FETs with an $R_{DS(on)}$ of 0.2. This was unavoidable as they were the only available FETs with a safe operating area that enclosed 12V 8A. It's possible that a larger heatsink would allow larger currents- however a better solution was to use lower $R_{DS(on)}$ FETs, as implemented in the "production" variant.

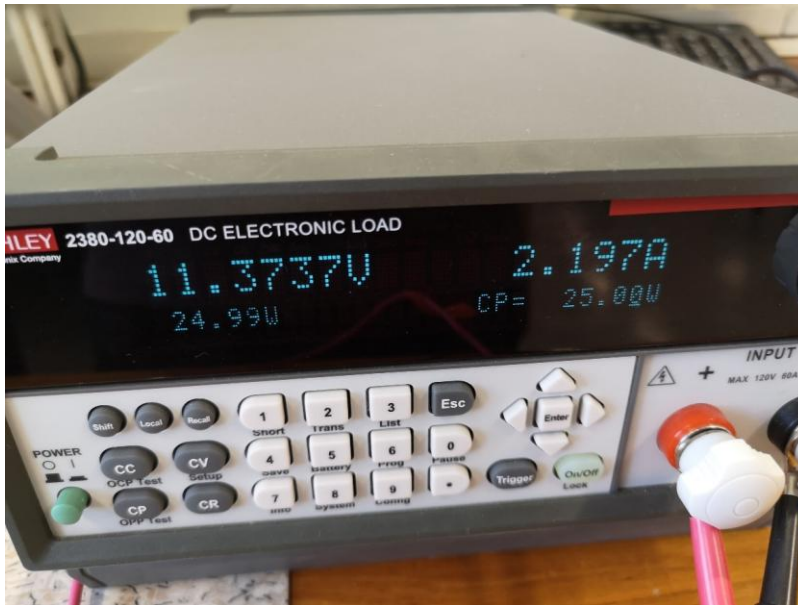


Figure 11: 25W Load Test Parameters

However 25W is still sufficient to illuminate a single incandescent bulb or complete string of LED bulbs, so being able to supply this proves the buck converter and load switch concepts- but not their implementation in the prototype.

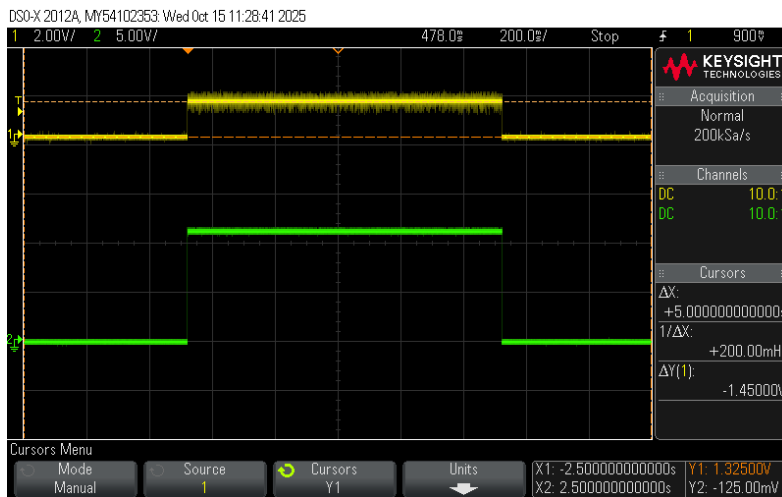


Figure 12: INA181 Output Voltage (Yellow) vs Voltage Across 4.72 Ohm Resistor (Green)

The response and accuracy of the current sensor was evaluated next. A known resistor measured at 4.72 Ohms was connected across one of the load-switch circuits, voltage was probed at the high side of this resistor and the output of the current sensor. When the load switch was turned on, the analog current output rose to 1.45 V. This signal was fairly noisy; however, this is likely due to improper grounding of the oscilloscope probe. An error of the designed PCB is that it doesn't have a grounded test point, so probe ground was instead connected to a jumper for testing- this probably led to exaggerated HF noise and ringing (Figure 14, 16). However, it could also be due to insufficient differential input filtering, this could be solved by increasing C3 which in turn would decrease the -3dB frequency, filtering more of the high frequency switching noise. It's unlikely to be due to power supply noise issues, although supplied upstream by the designed buck converter, 5V on the board is supplied by the AMS1117 on the Arduino board, which has a high ripple rejection ratio of 80 dB [5].

1.45 Volts is consistent with correct operation. With a gain of 20 and shunt resistance of 0.03, this yields a current of 2.41. Voltage across the resistor (Green trace) was measured to be 11.35, yielding a current of 2.40, which is very similar to the sensor measured current.

The buck converter output voltage was evaluated next. A 24 V supply voltage was connected, and varying loads were connected to a single load switch output channel. An oscilloscope probe was attached to the VOUT test point, and the output waveform was measured. No load (Figure 13), 12W (Figure 14) and 30W (Figure 15) data were collected. At no load, ripple was very high (835 mV P-P), this is because the LM25085 operates in discontinuous conduction mode under light loads [4]. This reduces the switching frequency to regulate voltage under low output current, from Figure 13 it switches at just 641 Hz under no load.

As load was increased, the controller switches to continuous conduction mode, where the on-time of the switch is varied while switching at a constant frequency. From the figure 4 and 5 seconds per division, this frequency can be determined to be 266 kHz, which is much higher than the designed 100 kHz. The switching frequency can be adjusted by varying RT (Figure X R25), in the prototype this was incorrectly set to 270k; manually calculating RT from datasheet equations gives a much higher 750k, which was used in the "production" schematic [4]. Additionally, large ringing artefacts can be seen when the MOSFET switches on at each ripple trough, and switches off near each ripple peak. The ringing could be due to the placement of the VOUT test point, which is immediately after the inductor and before the output capacitors physically.

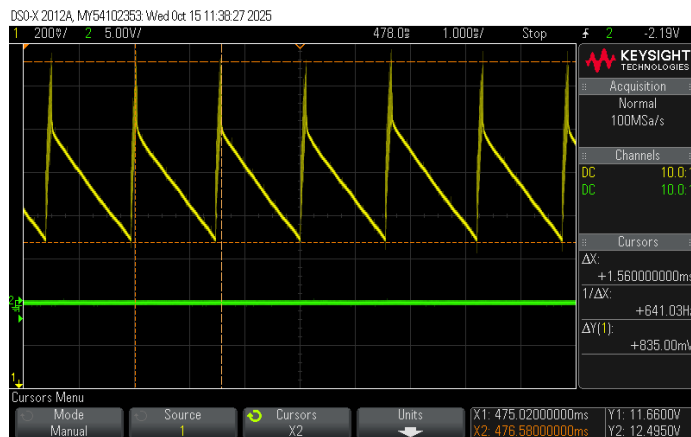


Figure 13: No Load Buck VOUT Waveform. Note 835 mV Ripple Voltage

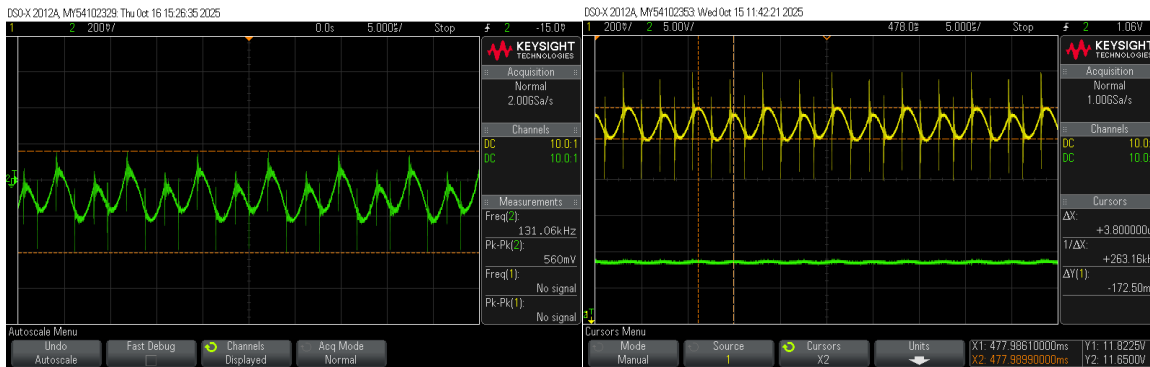


Figure 15: VOUT Waveform, 10 W Load
Figure 14: VOUT Waveform, 10 W Load

Figure 15: VOUT Waveform, 30 W Load

s due to the selection of feedback resistors. The 10k/1.2k ratio used in the prototype has a set-point voltage of around 11.7 rather than 12.0. The schematic was updated to a more accurate ratio. It can also be seen that the voltage ripple is very poor, at around 500 mV P-P at 10 W and 170 mV at 30 W. This could be due to the PCB layout, particularly the long, looping, feedback trace; further board developments could use interior layers to route this point-to-point, which would be impossible on a 2 layer board. It could also be due to output component selection, in particular the 10uH power inductor might not form a sufficient LC filter with the output capacitor selection. This could be fixed by using a larger inductor or adding more output capacitance. The feedback loop might not be compensated properly, C8 could be tuned to fix this. The large ripple and ringing should have little to no effect on operation, as automotive power supplies are much noisier.

To test the low battery cut off protection, our device was powered up using the lab bench supply. The input voltage was then gradually reduced from the nominal 18v. When the input reached 15v the device entered the LOWBATT state, shut off the lcd screen, and illuminated the red battery warning led. The test was considered successful. This battery protection does have limitations, in that there is no over voltage protection for the Arduino if an input is applied that is greater than 25 volts. To remedy this, a 5.1v zener diode could be placed in parallel with the 8.2k Ohm resistor in the voltage sensing divider. This would limit the voltage on the A0 pin to the zener voltage and provide a further layer of protection.

Throughout the testing process the software performed as specified. Once development was complete, no major bugs were found. Utilizing a state machine in software worked well and allowed for efficient debugging during development as we could easily see what state we were in. However, if future development of this product was to continue, it should be investigated whether an embedded systems approach is the right way to go. Could the logic of this state machine be implemented using hardware such as logic gates and would this be a more cost-effective approach.

6.2 Chassis

The final chassis successfully meets all functional/structural goals of the trailer lights tester. It provides a robust and rigid housing for all electrical components, including the PCB shield, Arduino, user interfaces, I²C LCD display, and can securely hold the Mikita battery adapter.

The chassis design also meets all requirements outlined in the specifications table (see Appendix A). The 130 mm x 130 mm size with 5mm riveted edges allows it to be handheld comfortably, and easily transportable, meeting S4.1, and allows it to be operated by one user, meeting requirement S5.1. The 3 mm walls of the chassis, and 4 mm thickness of the lid, along with the screws and display being flush with the lid, means it has a high durability, and robustness, allowing it to be used outdoors without risk of breaking. The chassis has holes for M10 cable glands, in both the side of the chassis, and the bottom, allowing both the trailer

cable, and power cables to enter both the side and bottom of the enclosure respectively, meeting requirement S4.3. The M4 screw heat set inserts at the bottom of the chassis allow the Mikita battery adapter to be securely fastened onto the bottom of the chassis, satisfying requirement S4.4. The cylindrical supports with M3 heat set inserts allows the PCB and Arduino to be securely fastened into the chassis reliability meeting requirement S4.5, and the use of heat set inserts in general, allow the chassis lid to be reliably opened frequently, without causing any damage to connections, along with the I²C LCD display, PCB, Arduino, and the Makita battery adapter. Securing the Makita battery adapter to the bottom of the chassis, allows a Makita 18V battery to slide onto and off of the chassis, meeting requirement S2.1, which allows efficient charging/replacement of battery while being used in the field.

Testing showed that all internal and external components mounted securely (see Figure 16-18), using the M3, and M4 heat set inserts. Repeated assembly was possible with no wear or damage of plastic. The M4 base heat set inserts anchored the Makita battery adapter strongly in place (see Figure 17). The M10 cable gland provided effective protection of wires leaving the chassis. Tolerances in design were verified in the assembly of the unit, all being within $\pm 0.5\text{mm}$.

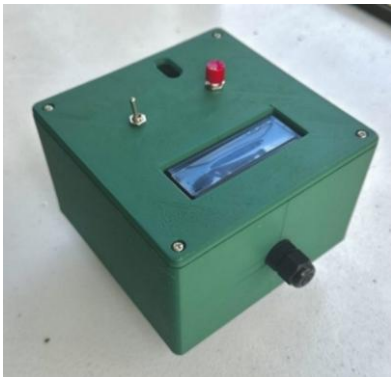


Figure 16: Full assembly (Top view)

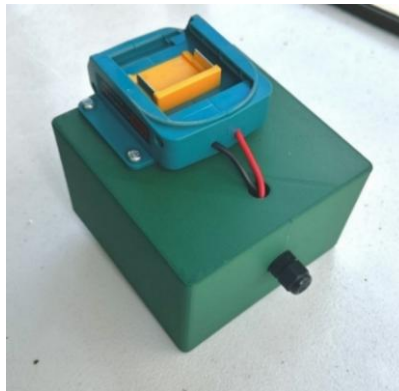


Figure 17: Full assembly (Bottom view)

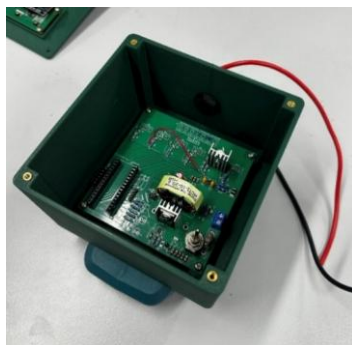


Figure 18: Full assembly (PCB View)

7. Conclusions

The trailer lights tester successfully achieved the goal of providing a portable single user device capable of verifying trailer lights without a towing vehicle. The design integrates efficient power regulation, reliable load control, intuitive software and user interface, and a durable enclosure.

Power from an 18 V Makita battery is converted to a stable 12 V using an LM25085 buck converter, ensuring safe operation of both LED and incandescent circuits. High-side P-channel MOSFETs provide efficient switching, while INA181 current-sense amplifiers enable accurate fault detection. The system is controlled by an Arduino Nano microcontroller programmed in C as a state machine, featuring single-button operation, LCD user feedback, automated fault detection, and undervoltage protection. The modular software structure also allows future expansion to support additional trailer configurations or power sources. All components are enclosed within a compact 3D-printed chassis designed for portability, durability, and ease of assembly.

The result is a robust, efficient, and maintainable trailer light tester that meets stakeholder requirements and demonstrates effective integration of electronics, software, and mechanical design.

8. References

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Commented [BW4]: Did you guys have the data sheet links, etc

Commented [5R4]: what data sheets should we include?

Commented [BW6R4]: I was mainly thinking of the TI schematic. Parts used on the PCB probably aren't necessary

Commented [BW7R4]: Nevermind bro has everything already sorted

Appendices

Appendix A: Table of Specifications

System Inputs & Outputs	
Inputs	<i>Momentary Push Button</i> : Button press to start test sequence, view errors, restart test <i>Power Switch</i> : Toggle switch to turn on device
Outputs	<i>Light Sequence</i> : The light sequence illuminated during the test so tester can verify functionality of lights <i>LCD Display</i> : The test results displayed, state of device, battery level <i>Power Indicator</i> : Green LED indicating Arduino is Powered
Behavior and Constraints	
ID	Specification
S1	Test Sequence
S1.1	Each light must be lit for 1 second, enough to verify light function, and be adequately visible
S1.2	0.5 second delay between each light, for balance between speed of test, and visibility of each individual light
S1.3	7 second delay from button to start the test, to the beginning of the sequence, giving user enough time to walk from connector to the back of the trailer
S1.4	Lights are bright enough to be visible for testing in daylight
S2	Power
S2.1	Battery can be powered by a Makita 18V battery, which can slide into the bottom of the chassis
S2.2	Battery can be easily disconnected and reconnected to testing unit, for recharging/replacement
S2.3	Cutoff 15V triggers red LED, which prevents further testing, indicates battery is low
S2.4	Device capable of testing at least 10 trailers per charge
S3	Display
S3.1	Display "Test Ready, press button to start"
S3.2	After concluding test - If no faults found, display "All tests successful on LCD display" - If some tests fail, each button press displays each failed test

S3.3	One long press takes the user back to the "Test Ready" state
S4	Chassis
S4.1	Chassis is compact handheld size, can be carried easily by one person
S4.2	Durable with thick walls, and can withstand outside conditions
S4.3	Chassis has room for connector wire to exit from the side, and power to enter through the bottom
S4.4	Battery adapter can be secured onto the chassis securely
S4.5	Chassis can hold Arduino Uno and PCB shield in place reliably, without movement
S4.6	Chassis can be opened repeatably and reliably, along with parts, for maintenance and repair without damage to parts
S5	Operation
S5.1	Single user able to operate device entirely
S5.2	Device can connect to seven-pin trailer plug, and trailer wire can fit into chassis

Appendix B: Code

```
#include <stdint.h>
#include <stdio.h>
#include <stdbool.h>
#include <Wire.h>
#include <hd44780.h>
#include <hd44780ioClass/hd44780_I2Cexp.h>

#define LED_ON HIGH
#define LED_OFF LOW

// Create LCD object
hd44780_I2Cexp lcd;

// each light is characterised by its pin number, fault state, and name
typedef struct {
    uint8_t pin;
    bool fault;
    char name[16];
    char power[16];
} Light;

// state machine states
typedef enum {
    LOWBATT,
    IDLE,
    RUN,
    RESULTS
} State;

//constants
//hardware configuration
const uint8_t NUM_PINS = 6;
const uint8_t POWER_LED = 8;
const uint8_t BATTERY_LED = 13;
const uint8_t PUSH_BUTTON = 4;

//set the timings for the lights and button in milliseconds
const unsigned long ON_INTERVAL_MS = 1000;
const unsigned long OFF_INTERVAL_MS = 500;
const unsigned long START_INTERVAL_MS = 7000;
```

```

const unsigned long LONG_PRESS_MS = 1500;
const unsigned long DEBOUNCE_MS = 50;

//ADC & scaling
const float FAULT_LOW_WATTS = 40.0;           //threshold for fault in watts
const float FAULT_HIGH_WATTS = 50.0;         //upper error threshold
const float ADC_REF_VOLTAGE = 5.0;
const float VOLTAGE_DIVIDER_GAIN = 5.024;
const float ADC_MAX_READING = 1023.0;
const float INA181_GAIN = 20.0;
const float SENSE_RESISTOR_OHMS = 0.03;
const float BATTERY_LOW_LIMIT = 15.0;
const unsigned long BAUD_RATE = 9600;

//set the I/O pins
Light trailer_lights[6] = {
  {2, false, "Right-hand turn", ""},
  {3, false, "Rear lamps", ""},
  {9, false, "Reversing signal", ""},
  {10, false, "Service brakes", ""},
  {11, false, "Stop lamps", ""},
  {12, false, "Left-hand turn", ""}
};

//declare remaining globals
uint8_t current_light;
uint8_t num_faults;
unsigned long previous_time;
bool start_test;
bool countdown;
bool release;
State current_state = IDLE; //FSM current state, start in IDLE
bool timer(const unsigned long interval);
void battery_level(void);
bool fault_detected(Light*); // main fault detection function
bool uno_fault_detected(void); // fault detection function used only for testing
with arduino uno on breadboard

void setup()
{
  Serial.begin(BAUD_RATE);

  //setup pins
  for (uint8_t i = 0; i < NUM_PINS; i++) {

```

```

        pinMode(trailer_lights[i].pin, OUTPUT);
        digitalWrite(trailer_lights[i].pin, LED_OFF);    // Give the output pins a
definite, known state at start up
    }
    pinMode(BATTERY_LED, OUTPUT);
    pinMode(POWER_LED, OUTPUT);
    pinMode(PUSH_BUTTON, INPUT_PULLUP);

    // check if battery voltage is > 15V
    battery_level();

    switch (current_state) {
    case LOWBATT:
        // battery <= 15V, remain in this state
        lcd.init();
        digitalWrite(BATTERY_LED, LED_ON);
        digitalWrite(POWER_LED, LED_OFF);
        lcd.off();
        break;
    default:
        //initialse variables, indicate ready state to user
        digitalWrite(BATTERY_LED, LED_OFF);
        digitalWrite(POWER_LED, LED_ON);
        lcd.init();
        lcd.backlight();
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Test Ready Press");
        lcd.setCursor(0, 1);
        lcd.print("Button to Start");
        previous_time = millis();
        start_test = false;
        current_light = 0;
        num_faults = 0;
        countdown = false;
        release = false;
        Serial.println("Setup complete");
        break;
    }
}

void loop()
{
    switch (current_state) {

```

```

case LOWBATT:
    // battery <= 15V, remain in this state
    lcd.off();
    digitalWrite(BATTERY_LED, LED_ON);
    digitalWrite(POWER_LED, LED_OFF);
    break;
case IDLE:
    //wait in this state until test run commanded by button press
    // wait until button released from previous state
    if (digitalRead(PUSH_BUTTON) == HIGH) {
        release = true;
    }
    //poll push button until pressed
    if (digitalRead(PUSH_BUTTON) == LOW && !start_test && release) {
        previous_time = millis();
        start_test = true;
    }
    //if button released trigger start of test
    if (digitalRead(PUSH_BUTTON) == HIGH && start_test && timer(DEBOUNCE_MS)) {
        previous_time = millis();
        Serial.println("Test in Progress");
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Test in Progress");
        start_test = false;
        countdown = true;
    }

    //start 7 second delay count, light first bulb, move to run state
    if (countdown) {
        if (timer(START_INTERVAL_MS)) {
            digitalWrite(trailer_lights[current_light].pin, LED_ON);
            current_state = RUN;
            countdown = false;
            release = false;
        }
    }
    break;
case RUN:
    //loop through each light, illuminating each bulb one at a time for the set
time
    if (digitalRead(trailer_lights[current_light].pin) == HIGH) {
        //take test data when light turned on

```

```

        trailer_lights[current_light].fault =
fault_detected(&trailer_lights[current_light]);

        //check time remaining on pin
        if (timer(ON_INTERVAL_MS)) {
            digitalWrite(trailer_lights[current_light].pin, LED_OFF);    //turn
off light
            current_light++; //move to next pin
        }
        //move to off state
    } else {
        if (timer(OFF_INTERVAL_MS) && current_light < NUM_PINS)
        {
            //check time remaining in off state
            digitalWrite(trailer_lights[current_light].pin, LED_ON);
//turn on light
        }
    }
    // if finished testing last pin reset the sequence and move to RESULTS
STATE
    if (current_light >= NUM_PINS) {
        current_light = 0;
        current_state = RESULTS;
        for (uint8_t i = 0; i < NUM_PINS; i++) {    // count number of
detected faults
            if (trailer_lights[i].fault) {
                num_faults++;
            }
        }
        if (num_faults != 0) {    //display detected faults
            lcd.clear();
            lcd.setCursor(0, 0);
            lcd.print(num_faults);
            lcd.print(" Faults Found");

        } else {
            lcd.clear();
            lcd.setCursor(0, 0);
            lcd.print("No Fault Found");

        }
    }
    break;
case RESULTS:

```

```

    // display the test results, if faults are found, user can cycle through
    each fault. Hold down button to return to IDLE
    //start timer to detect a long button press
    if (digitalRead(PUSH_BUTTON) == LOW && !start_test) {
        previous_time = millis();
        start_test = true;
    }

    //long press of button, return to IDLE state
    if (digitalRead(PUSH_BUTTON) == LOW && start_test && timer(LONG_PRESS_MS))
{
    Serial.println("Test Complete");
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Test Ready Press");
    lcd.setCursor(0, 1);
    lcd.print("Button to Start");
    current_state = IDLE;
    num_faults = 0;
    current_light = 0;
    start_test = false;
}

// short press of button
// if no faults detected return to IDLE
if (digitalRead(PUSH_BUTTON) == HIGH && start_test && timer(DEBOUNCE_MS)) {
    if (num_faults == 0) {
        Serial.println("Test Complete");
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Test Ready Press");
        lcd.setCursor(0, 1);
        lcd.print("Button to Start");
        current_state = IDLE;

        // if faults detected cycle through with each press
    } else {
        for (uint8_t i = current_light; i < NUM_PINS; i++) {
            if (trailer_lights[i].fault) {
                lcd.clear();
                lcd.setCursor(0, 0);
                lcd.print("Fault on P=");
                lcd.print(trailer_lights[i].power); //TEST
                lcd.setCursor(0, 1);
                lcd.print(trailer_lights[i].name);
            }
        }
    }
}

```

```

        previous_time = millis();
        current_light = i + 1;
        if (current_light >= NUM_PINS) {
            current_light = 0;
        }
        break;
    }
}

start_test = false;
}

break;
}
//check if battery runs low during operation
battery_level();
}

bool timer(const unsigned long interval)
{
    unsigned long current_millis = millis();
    if (current_millis - previous_time >= interval) {
        previous_time = current_millis;
        return true;
    } else {
        return false;
    }
}

void battery_level(void)
{
    uint16_t battery_sensor = analogRead(A0);
    Serial.print(battery_sensor);
    float voltage = battery_sensor * ((VOLTAGE_DIVIDER_GAIN * ADC_REF_VOLTAGE) /
ADC_MAX_READING);    //scale factor of voltage divider on board
    if (voltage <= BATTERY_LOW_LIMIT) {
        current_state = LOWBATT;
    }
}

//this function takes the data from the on board current sensor, calculates the
power drawn by the currently on circuit
//and compares to a fault threshold value

```

```
bool fault_detected(Light* currentLight)
{
    float senseVoltage = (analogRead(A7) * ADC_REF_VOLTAGE) / ADC_MAX_READING;
    float current = (senseVoltage / INA181_GAIN) / SENSE_RESISTOR_OHMS;    //
    INA181a1 has voltage gain of 20; sense resistor is 0.03 ohms
    float voltage = analogRead(A2) * ((VOLTAGE_DIVIDER_GAIN * ADC_REF_VOLTAGE) /
    ADC_MAX_READING); // measure voltage on output side of sense resistor for better
    accuracy
    float measuredPower = current * voltage;
    dtostrf(measuredPower, 0, 1, currentLight->power); // store measured power for
    fault display/normal display
    if (FAULT_LOW_WATTS > measuredPower || FAULT_HIGH_WATTS < measuredPower) {
        //Serial.println(currentLight->power);
        return true;
    } else {
        return false;
    }
}
```